

Yiliu (Aiden) Li

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Authorized to work in the U.S., U.K., and China; no visa sponsorship required.

Education

Northwestern University

M.S. in Computer Science (MSCS)

Expected Jun 2027

University College London

BSc Computer Science

Sep 2023 – Jul 2026

📄 Dissertation

Supervisor: Anthony Steed

Publications

- **Aiden Yiliu Li**, Nels Numan, and Anthony Steed*. “**Visual Agentic Memory: Enabling Online Long Video Understanding via Online Indexing, Hierarchical Memory, and Agentic Retrieval.**” 📄arXiv | 🔗Code
- Wee Joe Tan, Zi Rui Lucas Lim, Shashank Durgad, Karim Obegi, and **Aiden Yiliu Li***. “**Avenir-UX: Automated UX Evaluation via Simulated Human Web Interaction with GUI Grounding.**” 📄arXiv | 🔗Code
- Yiheng Tian, Jingnan Zhang, Yiven Zhu, Fei Xu, Hao Kan, Leenuja Kannan, Niyaa Meganathan, and **Aiden Yiliu Li***. “**From Advisor to Voting Teammate: Institutional Authority and Information Structures of AI Agents in Bounded-Rational Human Groups.**” *CHI 2026 Workshop*. 📄PDF
- **Aiden Yiliu Li**, Xinyue Hao, Shilong Liu*, and Mengdi Wang*. “**Avenir-Web: Human-Experience-Imitating Multimodal Web Agents with Mixture of Grounding Experts.**” 📄arXiv | 🔗Code
- **Aiden Yiliu Li**, Bizhi Yu, Daoan Lei, Tianhe Ren, and Shilong Liu*. “**Chain-of-Ground: Improving GUI Grounding via Iterative Reasoning and Reference Feedback.**” 📄arXiv
- Violet Yinuo Han, Ziru Wei, **Aiden Yiliu Li**, Chris Wu, and Alexandra Ion*. “**Embodiment and Interaction Influence Perceptions of Robotic Collaborators in Everyday Physical Tasks.**”
- Jing Xue, Madhankumar Vasudevan, **Aiden Yiliu Li**, Zhouyang Shen, James Hardwick, Martin Dechant, and Marianna Obrist*. “**KiTaFab: Kinaesthetic and Tactile Mid-air Haptics Fabric Experiences.**”

Research & Teaching Experiences

Research Internship

Stanford University Vision and Learning Lab (SVL)

Mar 2026 – Present

- Conducting research in Computer Vision and Machine Learning, focusing on physical intelligence and structured representations for visual reasoning.
- **Supervisor:** Prof. Jiajun Wu.
- **Mentor:** Yue Gao.

Research Internship

Princeton University AI for Accelerating Invention (AI2) Lab

Sep 2025 – Present

- Worked with Dr. Shilong Liu and Prof. Mengdi Wang on agentic AI, focusing on web-use agents and GUI-grounded interaction.
- Developed an open-source system achieving state-of-the-art performance among open-source methods, with work under submission to ICML.

Research Internship

Carnegie Mellon University Human Computer Interaction Institute (CMU HCII)

Jun 2025 – Sep 2025

- Conducted research in Physical Intelligence and Human-Robot Interaction by integrating VLM and VLA on robotic arms.
- Conducted research on how embodiment and interaction shape perceptions of robotic collaborators (submitted to CHI 2026).
- **Supervisor:** Prof. Alexandra Ion.

Research Assistant

Pervasive HCI Group, Tsinghua University

May 2025 – Dec 2025

- Conducting research in LLM Ideation and LLM Review by developing a GUI LLM platform to enable low-cost collection, organization, and synthesization of scientific facets.
- Designing and developing a cross-platform mobile web app for online AI hemifacial spasm analysis via facial analysis.

Research Assistant

UCL Computer Science Multi-Sensory Devices (MSD) Group

Aug 2024 – Apr 2025

- Developed and deployed a Virtual Reality (VR) environment that enhanced the experience of fabric touch perception.
- Contributed to research on kinaesthetic illusion through experimental support and literature review.
- Supported the preparation of presentations and research summaries about pseudo-haptics.
- **Supervisor:** Prof. Marianna Obrist.

Teaching Assistant

UCL COMP0016 System Engineering

Sep 2025 – Present

- Supervised multi-disciplinary, industry-sponsored CS teams (e.g., Microsoft, Intel, IBM, Cisco), owning weekly cadence,

stakeholder communication, and risk management to keep projects on track.

- Delivered deep technical mentorship across AI/ML (LLMs, NLP, RAG / Agentic AI), HCI, CG / AR / VR, computer vision, and cloud / mobile, assisting in system architecture and end-to-end implementation.
- **Module Leader:** Prof. Dean Mohamedally.

Teaching Assistant

Sep 2024 – Present

UCL ENGF0034 Design and Professional Skills I (Computer Science)

- Assisted students with Python coding, AI game auto player development, and debugging techniques.
- Provided support in understanding internet protocols and real-world networking applications.
- **Module Leader:** Prof. Martin Benning.

Industrial & Leadership Experiences

Head of Nexus Lab

Sep 2025 – Present

UCL Artificial Intelligence Society

- Directing Europe's largest student-led AI research lab, overseeing multi-pillar programmes and leading faculty / PhD-mentored teams from ideation to publication.
- Organized regular academic events to connect students and researchers, designed research roadmaps, and built structured pipelines guiding students from learning to active research participation.

Campus Ambassador

Oct 2025 – Present

TRAE, ByteDance

- Advocated for the AI-powered coding IDE on campus, organised workshops, and fostered a community of student developers.

Y Combinator AI Startup School

16–17 June, 2025

- Selected to join an exclusive program convening leading AI founders, researchers, and investors, including Sam Altman, Dr. Andrew Ng, Prof. Fei-Fei Li, Prof. Chelsea Finn, Garry Tan, and Satya Nadella, to explore frontier developments in AI innovation and entrepreneurship.

AI Product Manager Intern

Feb 2025 – Feb 2026

Ponder AI (ResearchFlow)

- Actively supporting initiatives such as Keyboard Shortcuts, Mind Map View, and Accuracy Optimization.
- Proposed, designed, and launched a multi-functional AI chatbox that supports in-contextual quotation.
- Assisted in structuring product roadmaps, gathering user feedback, and optimizing AI functionalities to enhance usability and user engagement.

Projects

Ageniti

- An SDK for building apps that can be called by agents.
- Defines an action once and exposes it across CLI, HTTP, MCP, OpenAI / AI SDK tools, React, and typed clients through a shared runtime.
- **Website:** ageniti.dev
- **GitHub:** github.com/Ageniti/ageniti

SightLinks – UCL IXN Project

Oct 2024 – Mar 2025

- Collaborated with UCL GDI Hub, Wheelchair Alliance, and Soundscape to enhance accessibility solutions for disabled individuals.
- Trained machine learning models based on MobileNetV3 and YOLO11, and used georeference techniques for precise zebra crossing detection and geolocation from satellite imagery, with precision above 85%.
- Achieved the **Highest Score in the Cohort**.
- **Project page:** students.cs.ucl.ac.uk/2024/group15

CUDA Accelerated SDOT Kernel Development at UCR Supercomputing Lab

Jun 2023 – Aug 2023

- Achieved a breakthrough in SDOT computation by optimizing GPU-accelerated kernels, enhancing efficiency, and minimizing data movement through CUDA programming and related optimization techniques.
- **Source code:** github.com/yiliu-li/Optimized-Cuda-SDOT-Kernel-on-NVIDIA-Turing-GPUs
- **Supervisor:** Prof. Zizhong (Jeffery) Chen.